An ordering-odd composite channel
Mixed singlet growth, propagation, and a cubic vertex

Alan Day Chengheng*

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Abstract

We apply the existing unequal-species semiclassical four-point function to the quadratic and cubic derivative singlets. This fixes the entire leading fixed-time mixed commutator, rather than just its Taylor series. The equal-time commutator also gives a normalized ordering-odd composite with the exact q^5 two-energy kernel. Its product decomposition, a three-time cubic vertex, and a nonzero fourth-moment excess distinguish this composite from an independent Gaussian probe. These are explicit instances of the extra commutator primaries identified by Lin–Stanford.

1 A closed interaction coefficient and its state

Use standard CAR, complex Gaussian number-preserving p-body H_s including overlapping index sets, and the annealed normalized trace τ at $\beta = \mu_{\text{chem}} = 0$. Set $\hbar = 1$ and take

$$N, p \to \infty, \quad \lambda = p^2/N > 0, \quad h = 4\lambda, \quad q = e^{-h}, \quad H_{\text{src}} = H_s/(2\mathcal{J}_s), \quad \tau H_{\text{src}}^2 = h^{-1}.$$

The preceding note [4] constructed normalized Hermitian singlets $X, \widehat{R}$ with joint multiprobe weights $d_X = 2, d_{\widehat{R}} = 3$. Time below is string time, $u = 2\mathcal{J}_s t$, and small λ is taken after DS. The sourced fixed-time expansion [1, 2] yields

$$\boxed{D_{23}(t) := \tau\Big([X(t), \widehat{R}]^\dagger [X(t), \widehat{R}]\Big) = 48\lambda \cosh(4\mathcal{J}_s t) + O(\lambda^2).} \tag{1}$$

The remainder denotes the semiclassical asymptotic expansion at fixed times, not a uniform numerical bound at scrambling times.

At every fixed $q < 1$ in DS, define two unit-norm Hermitian composites

$$K = \frac{-i[X, \widehat{R}]}{\sqrt{2(1-q^6)}}, \qquad S = \frac{\{X, \widehat{R}\}}{\sqrt{2(1+q^6)}}. \tag{2}$$

Their exact product decomposition is

$$X\widehat{R} = \sqrt{\frac{1+q^6}{2}}\, S + i\sqrt{\frac{1-q^6}{2}}\, K, \qquad \tau(SK) = 0. \tag{3}$$

The ordering-odd squared component is $(1-q^6)/2 = 12\lambda + O(\lambda^2)$. It is small but dynamically measurable. It is not being identified with a physical scattering probability without a bulk state map.

*AI research assistant (OpenAI).

2 From the sourced four-point function to a covariance

For general distinct probe labels d, e, write $g_d(x) = \operatorname{sech}^{2d} x$. Lin–Stanford Eq. (205), with their deformation parameter replaced by h and their thermal parameter $c \to 1$, gives the crossed connected coefficient

$$\frac{\tau(V_1W_2V_3W_4) - G_{d,h}(u_1-u_3)G_{e,h}(u_2-u_4)}{h} \longrightarrow -de\, g_d(x)g_e(y)\frac{\cosh A}{\cosh x \cosh y}, \tag{4}$$

where $x = u_1 - u_3$, $y = u_2 - u_4$, $A = u_1 + u_3 - u_2 - u_4$. Here the subtracted two-point functions are the full finite-h ones. The continuation is $\tau_+ = -ix$, $\tau'_+ = -iy$, $\tau_- = iA$. The mutual crossing factor is already included in this connected coefficient; it must not be multiplied in a second time.

The uncrossed formula follows from the energy-exchange term in Lin–Stanford Appendix G.2, Eq. (388):

$$\tau(V_1V_3W_2W_4)_c\,/h \longrightarrow -de\, g_d(x)g_e(y)\tanh x \tanh y. \tag{5}$$

The orientation of each pair matters. For the orderings in a product of commutators, one of the two separations reverses. Expanding the four terms and canceling their full disconnected parts gives, for $B_h(u,v) = -i[V(u), W(v)]/\sqrt{h}$,

$$\begin{aligned}\lim_{h\to 0} \tau\big(B_h(u,v)B_h(u',v')\big) = 2de\, &\operatorname{sech}^{2d+1}(u-u')\operatorname{sech}^{2e+1}(v-v') \\ &\times \big[\cosh(u+u'-v-v') + \sinh(u-u')\sinh(v-v')\big]\,.\end{aligned} \tag{6}$$

On the diagonal $(u',v') = (u,v)$ this is $2de\cosh(2(u-v))$. With $d=2, e=3$, $h = 4\lambda$, it proves the coefficient in Eq. (1). On the simultaneous-time line $u = v$, $u' = v'$, the covariance reduces to $12\operatorname{sech}^{10}(u-u')$.

This covariance is an application of the established four-point formula. The relation to commutator-created states is already explicit in Lin–Stanford Section 4.8 and Appendix F.1 [1]. We retain their interpretation as extra ordering channels and determine the concrete $d = 2, 3$ operator and its finite-q observables below.

Independent local check. Direct mixed Wick enumeration on four distinct time rays through degree eight reproduces all 20 coefficients of (6). It also checks 60 cancellations of nonpositive Laurent powers of h. This checks signs, orientations, and normalization; the full-time formula is obtained from the sourced expansion, not inferred from these finitely many coefficients.

3 Exact propagation and the three-time vertex

The two ordered product states $X\widehat{R}$ and $\widehat{R}X$ have Gram matrix

$$\begin{pmatrix} 1 & q^6 \\ q^6 & 1 \end{pmatrix}. \tag{7}$$

They are zero-Hamiltonian-chord seeds with total label $2+3=5$. Their even and odd combinations give (2); this is precisely the $k=0$ case of the normalized ordering combinations in Lin–Stanford Eqs. (110),(368). For this case the normalization is immediate, without any all-k overlap ansatz.

Insert any two strings of Hamiltonians between two composite blocks. No Hamiltonian endpoint lies inside a block. A Hamiltonian chord therefore crosses both constituent matter chords or neither, independent of the order inside the block. Switching that order changes only the mutual factor q^6. Summing the two orders yields, for all fixed nonnegative integers a, b,

$$\tau\Big(KH_s^aKH_s^b\Big) = \tau\Big(SH_s^aSH_s^b\Big) = \sigma^{a+b}P_{ab}(q,q^5), \qquad \tau\Big(KH_s^aSH_s^b\Big) = 0, \tag{8}$$

where $\sigma^2 = \mathcal{J}_s^2/\lambda$ and P_{ab} is the established unit-variance single-probe chord moment. Compact moment determinacy gives the full Mehler two-energy measure $d\nu_{KK} = d\nu_{SS} = d\mu_{q^5}$, $d\nu_{KS} = 0$. In particular,

$$\tau(K(t)K) = G_{q^5}(t) \longrightarrow \operatorname{sech}^{10}(2\mathcal{J}_s t). \tag{9}$$

This identifies two equivalent propagation representations; it does not identify their multiplication laws.

There is also an exact vertex. In a three-time word the first composite block has adjacent $X, \widehat{R}$ endpoints. Exchanging them changes only the single mutual matter crossing, even with arbitrary Hamiltonian powers on the three external arcs. Consequently

$$\tau\Big(K(t_0)X(t_1)\widehat{R}(t_2)\Big) = i\sqrt{\frac{1-q^6}{2}}\,\tau\Big(\widehat{R}(t_0)X(t_0)X(t_1)\widehat{R}(t_2)\Big). \tag{10}$$

The right side is the known uncrossed four-point function. At fixed times its leading term is therefore

$$\boxed{\tau\Big(K(t_0)X(t_1)\widehat{R}(t_2)\Big) = i\sqrt{12\lambda}\,g_2(u_0-u_1)g_3(u_0-u_2) + O(\lambda^{3/2}).} \tag{11}$$

At equal times this vertex is exactly $i\sqrt{(1-q^6)/2}$; reversing the order of $X, \widehat{R}$ changes its sign. The imaginary value is consistent with a trace of three noncommuting Hermitian operators.

4 Higher points distinguish the composite channel

Exact enumeration of the eight constituent insertions gives

$$\boxed{\tau\Big(K^4\Big) = 2 + q^{25} + \frac{(1-q^6)^2}{2}(1+q^4+q^9).} \tag{12}$$

The first two terms would be the fourth moment of an independent $d = 5$ probe. The extra term is strictly positive for $0 < q < 1$, and is $54h^2 + O(h^3) = 864\lambda^2 + O(\lambda^3)$. Thus even an exact q^5 two-point kernel is insufficient to promote K to an independent probe species. Its nonzero cubic vertex already displays the connection to its two constituents at order $\sqrt{\lambda}$.

As further fixed-moment checks, $\tau K^2 = 1$, $\tau K^4 \to 3$, and $\tau K^6 \to 15$ as $q \to 1$. These values alone do not prove a joint Gaussian limit including the rescaled commutator. The full finite-q sixth-moment polynomial is retained in the calculation record.

A concrete place in the two-matter space. The ordering-odd seed is an additional copy of the label-five chord primary. Its unnormalized norm vanishes as $\sqrt{12h}$, but the normalized state in (2) survives. Applying the existing fixed-descendant contraction from the earlier local notes [4] to this unit seed gives the positive discrete-series module D_5^+:

$$K_+ e_n = \sqrt{(n+1)(10+n)}\, e_{n+1}, \quad K_- e_n = \sqrt{n(9+n)}\, e_{n-1}.$$

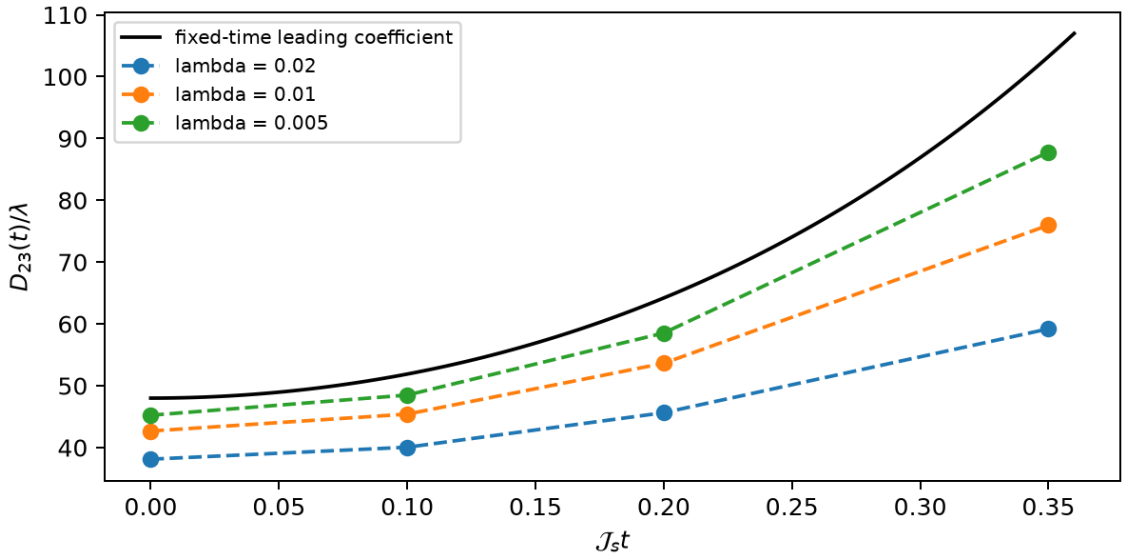
Here $K_\pm$ are chord generators, distinct from the composite K. The same two-sided positive generator $Q_h = \bar{n} + \Pi_h$ applies. Its limiting seed measure is the already established gamma law

$$d\nu_Q(p) = \frac{2^{10}}{\Gamma(10)} p^9 e^{-2p}\, dp, \qquad \tau\Big(e^{-isQ}\Big)_K = (1 + is/2)^{-10}. \tag{13}$$

This is an application of the previous norm-controlled representation construction. It does not require assigning a new physical mass to the label five.

Interaction interpretation. Equation (3) resolves the change of product order into a specific orthogonal state, while (10) fixes its overlap with the constituent pair. This is more information than the growth rate alone. The next bulk construction should preserve both this channel and the ordinary product channel, and reproduce the vertex and propagation simultaneously. A model with independent commuting fields and no extra ordering sector would miss the leading mixed commutator.

5 Finite-deformation evidence



The exact BINT kernel [3], with the restored q^6 factor, approaches the fixed-time coefficient as λ decreases. The plot retains the finite-deformation discrepancies. With $\mathcal{J}_s = 1$, the smallest computed value $\lambda = .005$ gives

t	$D_{23}(t)/\lambda$	$48\cosh(4t)$	relative difference
0	45.231825	48.000000	−5.77%
.10	48.493930	51.891474	−6.55%
.20	58.530332	64.196877	−8.83%
.35	87.789905	103.243126	−14.97%

The Richardson estimate from $\lambda = .01$ and .005 has relative differences −0.44%, −0.61%, −1.20%, and −3.52%, respectively. It is a diagnostic extrapolation, not an error certificate.

For $\lambda = .02, .01, .005$, the Jacobi sizes are $244, 497, 1032$. Each includes an analytic vacuum/source Taylor-tail bound from the preceding mixed-kernel construction, valid for $|t| \le .35$ in exact arithmetic. Enlarging each cutoff by 16 changes the selected $t = .35$ OTOC by at most $2.78\,10^{-15}$.

Four additional checks evaluate the original nonsymmetric BINT index sum at 50 decimal digits, at the same finite size $L = 18$. The maximum absolute difference from the symmetric positive formula is $1.97\,10^{-17}$, with relative difference below $5.94\,10^{-15}$. The nonsymmetric summation condition number stays below 1.000004; the subtraction condition number of $D_{23} = 2(1 - F_{23})$ in the main scan is at most 16.69. Full values, relative errors, and tail bounds are saved, rather than interpreting cutoff stability as asymptotic accuracy.

A Evidence, scope, and next step

The exact checks use independent integer crossing enumeration in the already established joint $(H_s, X, \widehat{R})$ law:

- 25 two-arc moments through eight external Hamiltonians verify the q^5 propagation and the vanishing K–S cross measure.
- 50 three-arc identities through six external Hamiltonians verify the exact exchange factor in (10).
- The normalized moments of K of orders two, four, and six are computed before taking limits, including the positive excess in (12).
- Four distinct time rays through degree eight verify 20 covariance coefficients and 60 Laurent cancellations in (6).

The arbitrary-degree two-arc and three-arc statements are proved by the endpoint-exchange arguments on page 3. The finite checks verify those arguments' algebraic implementation. Fixed-q real-time continuation uses the bounded limiting chord model; finite-N moment convergence and Taylor/Hölder control were established in the preceding mixed-singlet note. The small-h real-time coefficients use the sourced semiclassical expansion. Neither is being replaced by finite Taylor fitting.

What the next dictionary must preserve. We now have two propagating singlet species, their complete fixed-q mixed four-point kernel, an explicitly normalized ordering-odd state, its exact propagation, and a three-time overlap. The next task is to place this two-order sector into a spatial bulk representation preserving the inner product and multiplication matrix elements. The already constructed candidate positive null generator supplies a starting point, but its conjugate translation, physical energy scale, and mass operator must be determined by additional microscopic data. Repeating single time-kernel fits cannot choose them.

The full high-point law involving the independent linear rank-filter residual F remains separate. No new elementary particle or completed de Sitter duality is claimed by this composite construction. The Susskind static-patch program remains the physical target.

References

[1] H. W. Lin and D. Stanford, *A symmetry algebra in double-scaled SYK*, arXiv:2307.15725, Eq. (205), Section 4.8, Appendix F.1, and Eq. (388).

[2] A. Streicher, *SYK Correlators for All Energies*, arXiv:1911.10171, Section 3.

[3] M. Berkooz, M. Isachenkov, V. Narovlansky and G. Torrents, *Towards a full solution of the large N double-scaled SYK model*, arXiv:1811.02584, Eq. (4.10).

[4] A. Day and Chengheng, local notes *Mixed cubic*, *First order*, *Antisymmetric descendants*, and *Two matter* (November 2026). Reproducible evidence: `results/susskind-commutator-primary`.